Future directions $t = \epsilon$

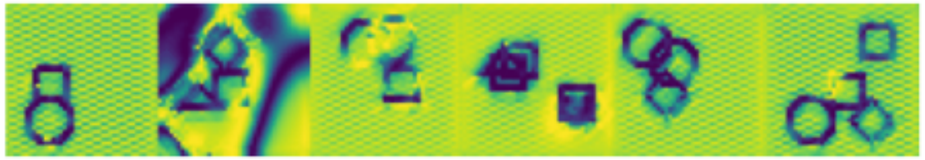
- **AdS-GNN+:** current model only uses scalar data, currently adding vector etc. data (results in better performance!)

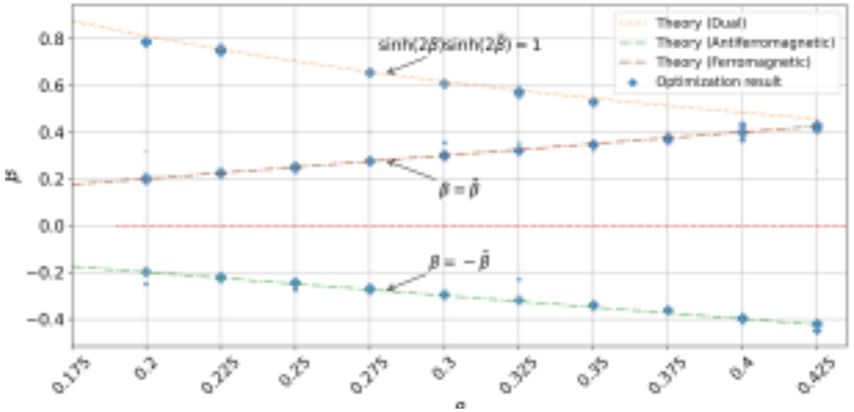
- **Goldstone modes for deep learning:** understanding Kuramoto-type models for deep learning (Miyato et. al 2024) from the point of view of Goldstone modes and spontaneous symmetry breaking. Work in progress with M. Welling, T. Miyato, V. Vitelli.

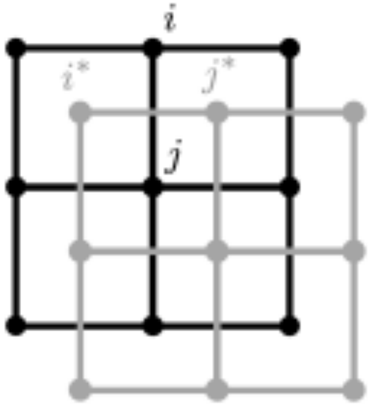
- **Finding dualities with ML**: Training neural networks to find duality mappings in statistical physics (with P. Gupta, A. Ferrari: NeurIPS 2024 Workshop ML4PS, under review ICML).

- **Learning cluster algorithms from diffusion:** Cluster algorithms dramatically enhance MCMC simulation at critical points, but are only known for specific models: learn them in generalusing ideas from generative AI. Work in progress with R. Bondesan, A. Hossain.

- **Ask me afterwards!**





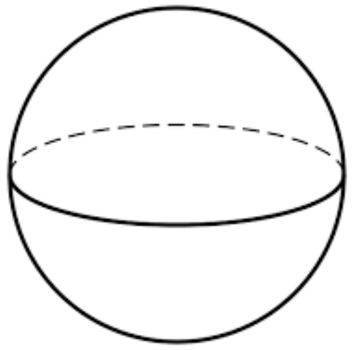


Future directions $t \to \infty$

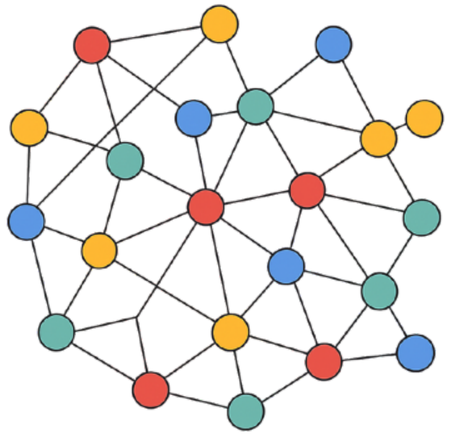
Neural networks are objects that can be fruitfully understood from principles used in physics.

Fluctuations

Symmetry



Locality



In physics these principles result in profound insights, e.g. the classification of phases of matter.

Machine learning gives us a new and fascinating class of systems to understand where **cherished physical principles** can find **new expression.** Exciting to see where these tools will take us.